

Unsupervised Learning

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1. Overleaf Read-Only Link

<https://www.overleaf.com/read/ykxmjhjhsjcj#8291b2>

2. GitHub Commit Hash

Repository: <https://github.gatech.edu/gt-omscs-ml/cs-7641-2026-spring-urafi3>

Final commit SHA: 884160cb424b6c0d81567621196100fe4abda1bd

3. Reproduction Instructions

a. Environment Setup

`pip install -r requirements.txt`

b. Data Setup

Expected directory layout:

```
ul/
├── main.py
├── requirements.txt
├── data/
│   ├── adult.csv
│   ├── wine.csv
├── output/
│   ├── figures/
│   └── results/
└── src/
    ├── __init__.py
    ├── paths.py
    ├── data_loader.py
    └── clustering.py
```

```

└─ dim_reduction.py
└─ neural_net.py
└─ step1.py
└─ step2.py
└─ step3.py
└─ step4.py
└─ step5.py

```

c. Random Seeds

All seeds are fixed in `src/paths.py` and passed explicitly to every scikit-learn and PyTorch object. No global numpy or torch seed is assumed.

```

RANDOM_STATE = 42      # train/test splits, KMeans init
N_INIT_KMEANS = 20     # KMeans independent restarts
N_INIT_GMM = 5         # GMM random initializations
RP_SEEDS = [0,1,2,3,4] # five RP runs for MSE variance

```

4. Running the Pipeline

The entire pipeline runs through a single entry point. All five steps execute in order automatically.

ul: the parent directory

python main.py: command to run the program

Figures are saved to `output/figures/` and result tables to `output/results/`. No command-line arguments are required.

No GPU is required.

5. EDA Summary Confirmation

Adult Income

The dataset contains 48,842 instances with 14 raw attributes and a binary income target ($\leq 50K$: 75.1%, $> 50K$: 24.9%). Missing values encoded as "?" in workclass, occupation, and native-country were imputed using the modal category per column. Categorical features were one-hot encoded yielding 108 binary columns; continuous features were standardized with StandardScaler. Capital-gain and capital-loss are heavily right-skewed, motivating StandardScaler over MinMaxScaler to avoid outlier scale domination.

Change from SL Report: None. EDA and preprocessing are identical.

Wine Quality

The dataset combines red ($n=1,599$) and white ($n=4,898$) Portuguese Vinho Verde samples for a combined $n=6,497$, with 11 physicochemical features and a discrete quality target rated 3–9 (seven classes). Classes

5 and 6 account for approximately 70% of samples. A binary color indicator was appended as a 12th feature. All features were scaled to [0,1] with MinMaxScaler. Notable pairwise correlations confirmed in EDA: free/total sulfur dioxide ($r \approx 0.72$), density/alcohol ($r \approx -0.69$), volatile acidity/quality ($r \approx -0.27$).

Change from SL Report: The binary wine color indicator was appended as a 12th feature for this assignment to allow clustering algorithms to leverage wine type information. This feature was not present in the SL Report feature set.